Unit 6: Cubic & Cube Root Functions Name: ____________________

Test Review: Transformations & Inverses Period: ________

Part 1: Quick Concepts

1. A function and its inverse are reflections across the line ____________.

2. If $f(7) = 20$, then $f^{-1}(20) =$ ________.

3. If $f(-5) = 12$, what point is on $f^{-1}(x)$? ____________

4. If $f(6) = -3$, what point is on $f(x)$? ____________

Part 2: Identify Parameters & Describe Transformations

5. $f(x) = -3(x+2)^3 - 4$

$a =$ ______ $h =$ ______ $k =$ ______

Transformations: ______________________

____

6. $f(x) = \frac{1}{2}\sqrt[3]{x-5} + 3$

$a =$ ______ $h =$ ______ $k =$ ______

Transformations: ______________________

____

Part 3: Write Equations from Descriptions

7. $f(x) = x^3$ is reflected over the x-axis, shifted left 4, and up 6.

$g(x) =$ ______________________________

8. $f(x) = \sqrt[3]{x}$ is vertically stretched by 2, shifted right 3, and down 1.

$g(x) =$ ______________________________

9. The function $g(x) = (x-c)^3 + d$ has its center point at $(-3, 7)$. Find c and d.

$c =$ ____________ $d =$ ____________

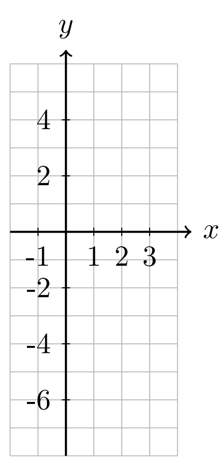
Part 4: Graph Transformations – Find center, transform points, graph, find intercepts.

10. $f(x) = 2(x-1)^3 - 4$

$a =$ _____ $h =$ _____ $k =$ _____

Center: (,)

Parent	**New**
$(-1,-1)$	(,)
$(0,0)$	(,)
$(1,1)$	(,)



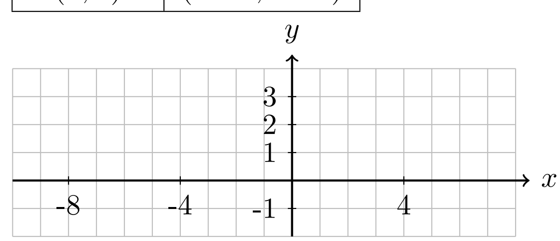
y-int: _________ x-int: _________

11. $f(x) = -\sqrt[3]{x+2} + 1$

$a =$ _____ $h =$ _____ $k =$ _____

Center: (,)

Parent	**New**
$(-1,-1)$	(,)
$(0,0)$	(,)
$(1,1)$	(,)



y-int: _________ x-int: _________

Part 5: Transform Specific Points

12. The parent function $f(x) = x^3$ contains the points $(1, 1)$ and $(-2, -8)$.
The transformed function is $g(x) = -3f(x-2) + 5 = -3(x-2)^3 + 5$.
Where would these points be located on $g(x)$?

$(1, 1) \to$ ____________ $(-2, -8) \to$ ____________

Find the x-intercept of $g(x)$: ____________

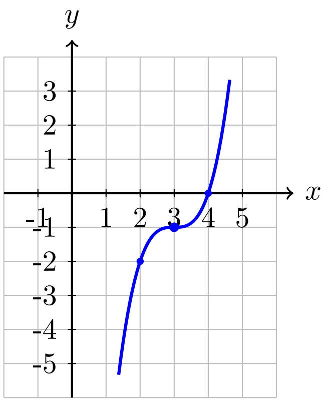
13. The parent function $f(x) = \sqrt[3]{x}$ contains the points $(8, 2)$ and $(-1, -1)$.
The transformed function is $g(x) = -f(x-4) + 2 = -\sqrt[3]{x-4} + 2$.
Where would these points be located on $g(x)$?

$(8, 2) \to$ ____________ $(-1, -1) \to$ ____________

Find the x-intercept of $g(x)$: ____________

Part 6: Write Equations from Graphs

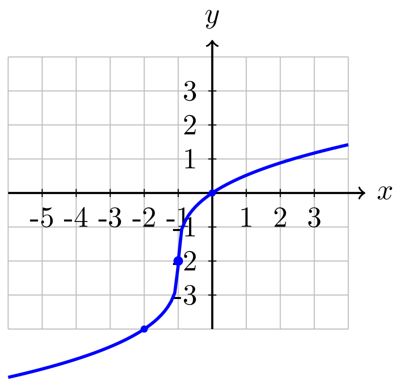
14.



Center: ________

$f(x) =$ ____________________

15.



Center: ________

$f(x) =$ ____________________

Part 7: Find Inverse Functions

16. Find the inverse of $f(x) = (x+4)^3 - 2$

$f^{-1}(x) =$ ____________________

17. Find the inverse of $f(x) = (2x-1)^3 + 3$

$f^{-1}(x) =$ ____________________

Part 8: Error Analysis

18. Marcus was asked to shift $f(x) = x^3$ up 4 units and right 2 units. He wrote $g(x) = (x+2)^3 + 4$. Explain his error and write the correct equation.

Correct equation: $g(x) =$ ______________________________

19. Destiny says $f(x) = 3(x-1)^3 + 2$ is vertically compressed. Find and correct her error.

Unit 6 Test Review (continued) Name: ________________________

Part 9: Multiple Choice Practice

20. Which represents $f(x) = x^3$ reflected over the x-axis and shifted left 5?

A. $g(x) = -x^3 + 5$ B. $g(x) = -x^3 - 5$ C. $g(x) = -(x-5)^3$ D. $g(x) = -(x+5)^3$

21. Given $h(x) = x^3$ and $p(x) = -2(x-3)^3 + 1$, which describes the transformations?

A. Vertical stretch by 2, reflect over x-axis, right 3, up 1
B. Vertical stretch by 2, reflect over x-axis, left 3, up 1
C. Vertical compress by $\frac{1}{2}$, reflect over x-axis, right 3, down 1
D. Vertical stretch by 2, right 3, down 1

22. In $f(x) = 4(x-2)^3 + 5$, which constant indicates a vertical stretch? __________
Which is a vertical shift?__________ Which is a horizontal Shift?__________

Part 10: Finding Intercepts of Transformed Functions

23. For the function $g(x) = 2(x+1)^3 - 16$:

a) Find the y-intercept by evaluating $g(0)$.

y-intercept: ______________

b) Find the x-intercept by setting $g(x) = 0$ and solving.

x-intercept: ______________

Part 11: Verify Inverses by Composition (Advanced Section)

Show that $f(f^{-1}(x)) = x$ OR $f^{-1}(f(x)) = x$ to verify the functions are inverses.

24. Verify that $f(x) = (x-1)^3 + 2$ and $f^{-1}(x) = \sqrt[3]{x-2} + 1$ are inverses by computing $f(f^{-1}(x))$.

25. Given $f(x) = (2x+3)^3 - 4$, find $f^{-1}(x)$, then verify by showing $f^{-1}(f(x)) = x$.

Quick Reference

Transformation Form:
Cubic: $f(x) = a(b(x-h))^3 + k$
Cube Root: $f(x) = a\sqrt[3]{b(x-h)} + k$

Parameters:

- $|a| > 1$: vertical stretch
- $|a| < 1$: vertical compression
- $a < 0$: reflection over x-axis
- $|b| > 1$: horizontal compression by $\frac{1}{b}$
- $|b| < 1$: horizontal stretch by $\frac{1}{b}$
- $h > 0$: shift right (opposite sign!)
- k: vertical shift (same sign)

Parent Anchor Points:

$y = x^3$	$y = \sqrt[3]{x}$
$(-2, -8)$	$(-8, -2)$
$(-1, -1)$	$(-1, -1)$
$(0, 0)$	$(0, 0)$
$(1, 1)$	$(1, 1)$
$(2, 8)$	$(8, 2)$

Point Rule: $(x, y) \to \left(\frac{x}{b} + h, ay + k\right)$
Inverse: swap x and y, solve for y